Operating Systems

Unit 4 – Orange Program Assignment

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SEC : 4 H

SCREENSHOTS:

1A)

shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs

```
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ nano object.c
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

1B)

```
shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

2A)

```
shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ nano tree.c
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

2B)

```
shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ find .pes/objects -type f
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/78/a314fb82bf73021279c1979372e752a91759746abe8aa9c1dda3cf4d1401be
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/c4/a9fb6638ffcac651e37e5ed4ccb987332ca99dc5340124cf19d467d2f88ba9
.pes/objects/d5/04d11677822b43da950d12118a21720fb92a5926aa8c052f77e0aeeaa2e20e
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
```

```
shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ xxd .pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6 | head -20
00000000: 7472 6565 2039 3600 3130 3036 3434 2062  tree 96.100644 b
00000010: 7965 2e74 7874 00e6 7ed6 6bcb 4a70 8250  ye.txt..~.k.Jp.P
00000020: a89f 315d 4d8b b927 0334 3c84 df83 4438  ..1]M..'.4<...D8
00000030: 880e 13a6 8652 bc31 3030 3634 3420 6865  .....R.100644 he
00000040: 6c6c 6f2e 7478 7400 05c8 d777 d0be 8574  llo.txt....w...t
00000050: 8b67 88d1 b5ca f7d6 9754 e6e6 8060 c265  .g.......T...`.e
00000060: ca21 cbd6 c05e 5f43                      .!...^_C
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

3A)

shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs

```
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ echo "hello" > a.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ echo "world" > b.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes add a.txt b.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes status
Staged changes:
  staged:     a.txt
  staged:     b.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  commit.h
  untracked:  file1.txt
  untracked:  index.c
  untracked:  index.h
  untracked:  Makefile
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h

shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

3B)

shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs

```
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776301376 6 a.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776301385 6 b.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

4A)

shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs

```
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ echo "Hello" > hello.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes add hello.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes commit -m "Initial commit"
Committed: d504d1167782... Initial commit
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ echo "World" >> hello.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes add hello.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes commit -m "Add world"
Committed: 78a314fb82bf... Add world
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ echo "Goodbye" > bye.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes add bye.txt
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes commit -m "Add farewell"
Committed: c4a9fb6638ff... Add farewell
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ ./pes log
commit c4a9fb6638ffcac651e37e5ed4ccb987332ca99dc5340124cf19d467d2f88ba9
Author: Shriya <PES1UG24CS496>
Date:   1776303598

    Add farewell

commit 78a314fb82bf73021279c1979372e752a91759746abe8aa9c1dda3cf4d1401be
Author: Shriya <PES1UG24CS496>
Date:   1776303566

    Add world

commit d504d11677822b43da950d12118a21720fb92a5926aa8c052f77e0aeeaa2e20e
Author: Shriya <PES1UG24CS496>
Date:   1776303534

    Initial commit

shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

4B)

shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs

```
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/78/a314fb82bf73021279c1979372e752a91759746abe8aa9c1dda3cf4d1401be
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/c4/a9fb6638ffcac651e37e5ed4ccb987332ca99dc5340124cf19d467d2f88ba9
.pes/objects/d5/04d11677822b43da950d12118a21720fb92a5926aa8c052f77e0aeeaa2e20e
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

4C)

shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs

```
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ cat .pes/refs/heads/main
c4a9fb6638ffcac651e37e5ed4ccb987332ca99dc5340124cf19d467d2f88ba9
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```

Branching and Checkout

Q5.1: A branch in Git is just a file in .git/refs/heads/ containing a commit hash. Creating a branch is creating a file. Given this, how would you implement pes checkout <branch> — what files need to change in .pes/, and what must happen to the working directory? What makes this operation complex?

• Update .pes/HEAD to point to refs/heads/<branch> and read the commit hash from that branch file.

• Load the commit object, extract its tree, and recursively reconstruct the working directory using blob objects.

• Create new files/directories, overwrite modified files, and delete files not present in the target branch to match the snapshot exactly.

• Complexity arises due to modifying the live filesystem, handling directory structures, permissions, and avoiding partial updates or conflicts.

Q5.2: When switching branches, the working directory must be updated to match the target branch's tree. If the user has uncommitted changes to a tracked file, and that file differs between branches, checkout must refuse. Describe how you would detect this "dirty working directory" conflict using only the index and the object store.

• Compare index metadata (size and mtime) with working directory files to detect possible modifications.

• If mismatch occurs, recompute the file hash and compare with the stored blob hash to confirm changes.

• Check the target branch’s tree for the same file path and compare blob hashes.

• If both local changes and branch differences exist, abort checkout to prevent overwriting uncommitted work.

Q5.3: "Detached HEAD" means HEAD contains a commit hash directly instead of a branch reference. What happens if you make commits in this state? How could a user recover those commits?

• Detached HEAD occurs when .pes/HEAD stores a commit hash instead of referencing a branch.

• Commits created in this state are valid but not pointed to by any branch.

• These commits may become unreachable if the user switches branches.

• Recovery is possible by creating a new branch pointing to that commit hash.

Garbage Collection and Space Reclamation

Q6.1: Over time, the object store accumulates unreachable objects — blobs, trees, or commits that no branch points to (directly or transitively). Describe an algorithm to find and delete these objects. What data structure would you use to track "reachable" hashes efficiently? For a repository with 100,000 commits and 50 branches, estimate how many objects you'd need to visit.

• Start from all branch heads and HEAD, and traverse all reachable commits using parent links.
• For each commit, mark its tree and recursively mark all blobs and subtrees referenced.
• Use a hash set to efficiently track all reachable object hashes.
• Delete objects not in the reachable set; in large repositories (e.g., 100,000 commits), total object visits can reach hundreds of thousands to a few million, depending on tree and blob sharing.

Q6.2: Why is it dangerous to run garbage collection concurrently with a commit operation? Describe a race condition where GC could delete an object that a concurrent commit is about to reference. How does Git's real GC avoid this

• During commit creation, objects are written before references are updated.

• GC may run simultaneously and treat these new objects as unreachable.

• This can lead to deletion of objects that a commit is about to reference, causing corruption.

• Prevent using atomic updates, safe commit ordering, and delaying deletion of recently created objects.

FINAL:

```
shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$ make test-integration
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 7e78d1c0398f... Initial commit

Log after first commit:
commit 7e78d1c0398f7511c8db8767a9b90c8d522003d06cf306ba7332cb47fd84db99
Author: Shriya <PES1UG24CS496>
Date:   1776305317

    Initial commit


--- Second Commit ---
Committed: d883b35db8d7... Update file.txt

--- Third Commit ---
Committed: a454fd80920d... Add farewell
```

shriya@ShriyaCRao: /mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs

```
--- Full History ---
commit a454fd80920d8fa38e4e876e5a2f57d6e0e6722232df4123bb7e4656230d5ab5
Author: Shriya <PES1UG24CS496>
Date:   1776305318

    Add farewell

commit d883b35db8d774b1cd440656a9e919ae3fdbedb61bed35a2b250872487f0d4d9
Author: Shriya <PES1UG24CS496>
Date:   1776305318

    Update file.txt

commit 7e78d1c0398f7511c8db8767a9b90c8d522003d06cf306ba7332cb47fd84db99
Author: Shriya <PES1UG24CS496>
Date:   1776305317

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
a454fd80920d8fa38e4e876e5a2f57d6e0e6722232df4123bb7e4656230d5ab5

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/7e/78d1c0398f7511c8db8767a9b90c8d522003d06cf306ba7332cb47fd84db99
.pes/objects/a4/54fd80920d8fa38e4e876e5a2f57d6e0e6722232df4123bb7e4656230d5ab5
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/d8/83b35db8d774b1cd440656a9e919ae3fdbedb61bed35a2b250872487f0d4d9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
shriya@ShriyaCRao:/mnt/c/Users/shriy/OneDrive/Desktop/PES2UG24CS496-pes-vcs$
```